

References and Resources for IBL Teaching

Start here for an overview of students' learning difficulties.

1. Weber, K. "Research Sampler 8: Students' difficulties with proof" [Link](#)

Articles highlighting students' difficulties with learning and learning beliefs that hinder their own learning in passive learning environments.

2. Dreyfus, T. (1999). Why Johnny Can't Prove. *Educational Studies in Mathematics*, 38(1/3), 85–109. https://doi.org/10.1007/978-94-017-1584-3_5
3. Harel, G. and Sowder, L. "Students' Proof Schemes: results from exploratory studies." In A. Schoenfeld, J. Kaput and E. Dubinsky (Eds.), *Research on Collegiate Mathematics Education*, Vol.3, M.M.A. and A.M.S. 1998, 234-283.
4. Muis, K. 2004. Personal epistemology and mathematics: A critical review and synthesis. <https://doi.org/10.3102/00346543074003317>
5. Schoenfeld, A. H. (1988). When Good Teaching Leads to Bad Results: The Disasters of "Well-Taught" Mathematics Courses. *Educational Psychologist*, 23(2), 145–166. https://doi.org/10.1207/s15326985ep2302_5
6. Selden, J., & Selden, A., Even good calculus students can't solve nonroutine problems. *Research Issues in Undergraduate Mathematics Learning Preliminary Analyses and Results* (pp.19-26) Edition: MAA Notes No. 33 Publisher: Mathematical Association of America Editors: J. Kaput, E. Dubinsky
7. Selden, J., & Selden, A. (1995). Unpacking the Logic of Mathematical Statements. *Educational Studies in Mathematics*, 29(2), 123–151. <https://doi.org/10.1007/BF01274210>
8. Selden, A., & Selden, J. (2003). Validations of Proofs Considered as Texts: Can Undergraduates Tell Whether an Argument Proves a Theorem? *Journal for Research in Mathematics Education*, 34(1), 4–36. <https://doi.org/10.2307/30034698>
9. Tall, D. (1994). Cognitive difficulties in learning analysis. Report on the teaching of analysis. A. Barnard. Weber, K. (2001). Student Difficulty in Constructing Proofs: The Need for Strategic Knowledge. *Educational Studies in Mathematics*, 48(1), 101–119.
10. Weber, K., & Lara Alcock. (2004). Semantic and Syntactic Proof Productions. *Educational Studies in Mathematics*, 56(2/3), 209–234. <https://doi.org/10.1023/B:EDUC.0000040410.57253.a1>

Some resources on the social outcomes of non-inclusive, non-student-centered teaching and learning environments, and the negative impacts on marginalized people.

11. AMS Survey 2017-18 of new doctoral recipients [Link](#)
12. Ernest, J. B., Reinholz, D. L., & Shah, N. (2019). Hidden competence: women's mathematical participation in public and private classroom spaces. *Educational Studies in Mathematics*, 102(2), 153–172. <https://doi.org/10.1007/s10649-019-09910-w>

13. Ellis, J., Fosdick, B. K., & Rasmussen, C. (2016). Women 1.5 Times More Likely to Leave STEM Pipeline after Calculus Compared to Men: Lack of Mathematical Confidence a Potential Culprit. *PloS One*, 11(7), e0157447–e0157447. <https://doi.org/10.1371/journal.pone.0157447>
14. Seymour, E., & Hunter, A.-B. (2019). *Talking about Leaving Revisited Persistence, Relocation, and Loss in Undergraduate STEM Education* (E. Seymour & A.-B. Hunter, Eds.; 1st ed. 2019.). Springer International Publishing. <https://doi.org/10.1007/978-3-030-25304-2>, [UofT Library](#)

The Four Pillars of IBL are (1) Students deeply engaged in rich mathematics, (2) regular opportunities for collaboration, (3) Instructor inquiry into student thinking, (4) Instructor focus on equity. **Evidence for IBL**, including leveling the playing field for women and people of color in Math.

15. Laursen, S. L., & Rasmussen, C. (2019). I on the Prize: Inquiry Approaches in Undergraduate Mathematics. *International Journal of Research in Undergraduate Mathematics Education*, 5(1), 129–146. <https://doi.org/10.1007/s40753-019-00085-6>
16. Conference Board of the Mathematical Sciences (CBMS) Active Learning Statement. <https://www.cbmsweb.org/2016/07/active-learning-in-post-secondary-mathematics-education/>
17. Freeman, et al. (2014). Active learning increases student performance in science, engineering, and mathematics. <https://www.pnas.org/doi/10.1073/pnas.1319030111>
18. Hayward, C., & Laursen, S. (2014). *Collaborative Research: Research, Dissemination, and Faculty Development of Inquiry-Based Learning (IBL) Methods in the Teaching and Learning of Mathematics; Cumulative Evaluation Report: 2010-2013*. [Report to the National Science Foundation] Ethnography & Evaluation Research, University of Colorado Boulder. [Executive summary Full report](#)
19. Laursen, S. L. (2013). [From innovation to implementation: Multi-institution pedagogical reform in undergraduate mathematics](#). In D. King, B. Loch, L. Rylands (Eds.), [Proceedings of the 9th DELTA conference on the teaching and learning of undergraduate mathematics and statistics](#), Kiama, New South Wales, Australia, 24-29 November 2013. Sydney: University of Western Sydney, School of Computing, Engineering and Mathematics, on behalf of the International Delta Steering Committee.
20. Laursen, S. L., Hassi, M.-L., Kogan, M., & Weston, T. J. (2014). Benefits for Women and Men of Inquiry-Based Learning in College Mathematics: A Multi-Institution Study. *Journal for Research in Mathematics Education*, 45(4), 406–418. <https://doi.org/10.5951/jresmetheduc.45.4.0406>
21. Smith, J. C. (2006). A sense-making approach to proof: Strategies of students in traditional and problem-based number theory courses. *The Journal of Mathematical Behavior*, 25(1), 73–90. <https://doi.org/10.1016/j.jmathb.2005.11.005>
22. Theobald et al, (2020). Active learning narrows achievement gaps for underrepresented students in undergraduate science, technology, engineering, and math. <https://www.pnas.org/doi/10.1073/pnas.1916903117>

Other resources and references

23. Ambrose, S. A. (2010). How learning works seven research-based principles for smart teaching (1st ed.). Jossey-Bass. [Link to UofT Library](#)
24. [MAA Instructional Practices Guide](#)
25. [IBL Montage Video](#) (32 minutes, 4 different examples)
26. What to do on day 1? [Setting the Stage](#), by Dana Ernst.
27. [The IBL Blog Playlist](#), SY for a set of articles on IBL teaching.
28. Learn more about IBL at the [Academy of Inquiry Based Learning](#)
29. Yoshinobu, S., & Jones, M. G. (2012). The Coverage Issue. PRIMUS : Problems, Resources, and Issues in Mathematics Undergraduate Studies, 22(4), 303–316. <https://doi.org/10.1080/10511970.2010.507622> (For those worried about coverage.)
30. *Mathematical microaggressions* by Francis Su (2 pages) is a short article about the words we do and the harm that can be done by things we say. [Link to article](#)